

Human Learning in Dynamic Environments: Fixed vs. Adaptive RL Models

Franka Böckmann

Ewa Godlewska

Iheb Marouani

David Raul Carranza Navarrete

Julian Calvin Rill

Introduction

What is the phenomenon we want to model?

We are interested in modeling how participants improve their choices over time as they learn which bandit is more rewarding. The objective is to model how individuals learn and adapt to changing reward contingencies in a dynamic decision-making environment (learning curves).

We aim to capture trial-by-trial learning and behavioral adaptation in a two-armed bandit task with a reversal learning paradigm.

By comparing two different reinforcement learning (RL) models, we seek to determine whether learning is best explained by fixed learning rates (Rescorla, & Wagner, 1972) or adaptive learning mechanisms that account for uncertainty (Gershman, 2015).

Why is that phenomenon relevant for understanding human cognition?

Understanding how individuals learn and adapt to changing reward contingencies in a dynamic decision-making environment within a reversal learning paradigm provides insight into cognitive flexibility, reinforcement learning, and decision-making under uncertainty (Cools, Clark, Owen, & Robbins, 2002).

The ability to detect environmental changes and adjust behavior accordingly is a fundamental cognitive process involved in real-world learning and adaptation.

Methods

Why is this modeling method appropriate for understanding the phenomenon?

Reinforcement learning modeling provides a trial-by-trial framework to understand how individuals learn from reward feedback and adapt to changes in a reversal-learning task (Cools, Clark, Owen, & Robbins, 2002). These models are well-suited for studying adaptive decision-making under uncertainty, as they simulate how agents gradually learn from experience.

Since the task involves a reward contingency shift (i.e. 80% vs 20% to 20% vs 80%), RL models can help explain

how participants detect and adapt to environmental changes. Moreover, comparing model-free RL, like Rescorla-Wagner, with Bayesian RL allows us to examine whether learning is more likely to be driven by fixed, error-based updates or by dynamically adjusting beliefs based on uncertainty.

The Rescorla-Wagner Model (RWM) captures incremental learning using a fixed learning rate, while the Bayesian Reinforcement Learning Model (BRM) extends this by incorporating uncertainty tracking and adaptive learning through the hazard rate.

This modeling approach also provides a framework for parameter estimation, where key cognitive parameters such as learning rate (α), inverse temperature (β), and hazard rate (λ) can be inferred from behavior.

These parameters reveal individual differences in how quickly new information is learned, how deterministically choices are made, and how rapidly past beliefs are updated in response to change. By fitting these models to behavioral data, we can gain deeper insight into the learning mechanisms driving human decision-making.

Which hypothesis/hypotheses do we seek to test by contrasting the models?

The primary hypothesis we seek to test by comparing the RWM and the BRM is related to the question of which learning model best explains participants' learning curves, or how participants improve their choices over time as they learn which bandit is more rewarding. Are they best explained by a fixed learning rate (RWM) or by an adaptive, uncertainty-driven process (BRM)?

Specifically, if individuals learn with a constant update rate, the RWM should provide a better fit. However, if individuals adjust their learning rate dynamically based on uncertainty, then the BRM, which incorporates a hazard rate for adaptive belief updating, should better explain behavior. In short:

- **H1:** If individuals learn with a fixed learning rate, the RWM will provide a better fit to human behavior, reflecting model-free reinforcement learning driven by prediction errors.
- **H2:** If individuals adjust their learning dynamically

based on uncertainty, the BRM will provide a better fit to human behavior, as it incorporates an adaptive hazard rate to account for belief updating in volatile environments.

Description of computational model(s)

What are the inputs, system properties, and outputs of your models?

The inputs to both models are the participant's action choices and the reward outcomes received on each trial.

The RWM explicitly tracks Q-values, which represent the expected reward for each action and are updated using a fixed learning rate. In contrast, the BRM maintains probabilistic beliefs over reward probabilities, which can be thought of as belief-based value estimates because they incorporate uncertainty into the learning process.

The outputs of the models differ in how they represent learning updates. The RWM outputs action selection probabilities based on updated Q-values, while the BRM outputs updated posterior distributions over reward probabilities for reflecting belief-based learning.

Additionally, both models include key parameters that characterize the learning process, such as the learning rate (α), inverse temperature (β), and, in the case of the BRM, the hazard rate (λ). These parameters are estimated by fitting the models to behavioral data.

Which assumptions does each model make?

The RWM assumes that learning follows a simple prediction error mechanism, where a fixed learning rate (α) determines how much the agent updates Q-values based on rewards. It assumes that past rewards influence future decisions proportionally to this learning rate and that action selection follows a softmax decision rule governed by an inverse temperature parameter (β), which controls the balance between exploration and exploitation.

The BRM, on the other hand, assumes that individuals track uncertainty dynamically, updating beliefs over reward probabilities using Bayesian inference. Instead of a fixed learning rate, it uses a hazard rate (λ), which controls how quickly the model adjusts to new information when a change in the environment is detected. This allows the model to be more adaptive, adjusting learning speed based on uncertainty rather than using a fixed update rule.

Describe the computational implementation of each model (e.g., model formulas)

The RWM updates Q-values following the equation:

$$Q_{t+1}(a) = Q_t(a) + \alpha(R_t - Q_t(a)) \quad (1)$$

where α is the learning rate, which determines the step size of updates, and R_t is the observed reward. The probability of

selecting an action follows a softmax function:

$$P(a) = \frac{e^{\beta Q(a)}}{\sum_{a'} e^{\beta Q(a')}} \quad (2)$$

where β is the inverse temperature parameter, controlling the balance between exploration (choosing randomly) and exploitation (choosing based on expected value). To model reversal learning, we implement a switch in reward probabilities at trial 50: Bandit 1 drops from 80% to 20% and Bandit 2 increases from 20% to 80%. This allows the model to capture adaptation to changing reward contingencies.

The BRM represents reward beliefs using Beta distributions, where each action a has parameters $\alpha_t(a)$ and $\beta_t(a)$, which store the history of successes (rewards received) and failures (non-rewards received). The expected reward probability (posterior mean) for each action is:

$$\mu_t(a) = \frac{\alpha_t(a)}{\alpha_t(a) + \beta_t(a)} \quad (3)$$

and action selection is based on a softmax decision rule:

$$P(a) = \frac{e^{\beta \mu_t(a)}}{\sum_{a'} e^{\beta \mu_t(a')}} \quad (4)$$

Unlike the RWM, which updates Q-values with a fixed learning rate, BRM adjusts its learning dynamically by incorporating a hazard rate (λ), which controls the balance between updating beliefs with new information and forgetting old experiences. The update equations for the Beta distribution parameters are:

$$\alpha_{t+1}(a) = (1 - \lambda)(\alpha_t(a) + R_t) + \lambda \cdot \alpha_0 \quad (5)$$

$$\beta_{t+1}(a) = (1 - \lambda)(\beta_t(a) + (1 - R_t)) + \lambda \cdot \beta_0 \quad (6)$$

where α_0 and β_0 are the prior values of the Beta distribution. The hazard rate λ determines the rate of forgetting, allowing the model to rapidly adapt when reward contingencies change.

Unlike RWM (fixed Q-values), BRM dynamically tracks uncertainty and allows for adaptive learning, making it more responsive to environmental changes such as reward reversals.

Description of the experiment

Provide an overview of the experiment. What are the independent variables and dependent variables of the experiment?

The experiment investigates how individuals learn and adapt to changing reward contingencies in a two-armed bandit task with a reversal learning paradigm. Participants are asked to repeatedly choose between two options (bandits), each offering a probabilistic reward. Initially, one bandit has a higher probability of reward (i.e., 80%) while the other has

a lower probability (i.e., 20%). At a predefined trial (i.e., trial 50), the reward contingencies reverse. The objective is to measure how participants detect and adapt to this change in reward structure.

The independent variables include:

- **Reward probabilities:** The reward probabilities of the two bandits, which initially differ (80%-20%) and reverse at trial 50.
- **Trial number:** Pre-reversal versus post-reversal, which separates the phases of the task.

The dependent variables of the experiment include:

- **Choice behavior:** The probability with which participants select each bandit on each trial.
- **Learning dynamics:** The rate at which participants adjust their behavior after the reward reversal.
- **Model parameters:** Estimated from behavioral data, including:
 1. **Learning rate (α):** for the RWM.
 2. **Hazard rate (λ):** for the BRM.
 3. **Inverse temperature (β):** that governs exploration-exploitation trade-offs.
- **Model fit:** The fit of each model to the observed participant data, assessed using metrics such as log-likelihood and BIC.

How much data were collected (number of participants and trials)?

The dataset consists of 11 participants, each completing 100 sequential trials in the Two-Armed Bandit Task.

Model simulation

How was the process of simulating data from the models?

Each model was initialized with parameter values drawn from prior literature. For the RWM, we used $\alpha = 0.1$ and $\beta = 2.0$ to reflect moderate learning and stochasticity (Erev, Ert, & Roth 2010). The BRM was initialized with uninformative priors ($\alpha_0 = 1$, $\beta_0 = 1$; Gershman, 2015), an inverse temperature of $\beta = 3.0$ (Daw, Gershman, Seymour, Dayan, & Dolan, 2011), and a hazard rate of $\lambda = 0.15$ to allow adaptive learning (Nassar, Wilson, Heasly, & Gold, 2010).

Simulated data were generated over 100 trials, with a reversal in reward probabilities at trial 50. On each trial, both models computed choice probabilities using a softmax function: the RWM used current Q-values, while the BRM used posterior means from the Beta distributions. An action was selected probabilistically, and a reward was delivered via a Bernoulli trial based on the reward probability of the chosen action.

After each trial, models updated their internal estimates. The RWM applied a prediction error rule using a fixed learning rate, while the BRM updated its Beta parameters via Bayesian inference, with the hazard rate controlling belief decay and sensitivity to change.

This process produced trial-wise simulated data, including choices, rewards, and internal model states, suitable for evaluating model recovery and behavioral predictions.

Model fitting

The process of fitting the model(s) to the data. and the preprocessing steps of the data.

We fit both the Rescorla-Wagner model and Bayesian reinforcement learning model to participant data using maximum likelihood estimation (MLE), identifying the parameters that best explained how participants learned and made decisions over time. Before fitting, we removed incomplete trials, normalized trial numbers across participants, and encoded choices numerically (Bandit 1 = 0, Bandit 2 = 1).

For the RWM, we defined a parameter grid with learning rates (α) ranging from 0.05 to 0.8 and inverse temperatures (β) from 0.1 to 10, reflecting plausible cognitive ranges. Each (α , β) pair was evaluated using a grid search: a RWM agent simulated trial-by-trial choices, computed the log-likelihood of observed data, and updated Q-values using the standard prediction error rule. The best-fitting parameters were those maximizing log-likelihood, and a heatmap visualized the fit landscape.

For the BRM, we defined a grid over inverse temperature (β) values from 0.1 to 10 and hazard rates (λ) from 0.01 to 0.5, with fixed uninformative priors ($\alpha_0 = 1$, $\beta_0 = 1$). A BRM agent computed softmax choice probabilities from posterior means and updated beliefs using Bayesian inference modulated by the hazard rate. Each parameter pair was evaluated via grid search, and the best-fitting parameters were selected based on log-likelihood. A heatmap again visualized the log-likelihood surface.

Parameter recovery

Describe how you performed parameter recovery for your models.

First, a parameter space was defined for each model. For the RWM, the learning rate (α) ranged from 0.01 to 0.8 and the inverse temperature (β) from 0.01 to 8. For the BRM, β ranged from 0.1 to 10 and the hazard rate (λ) from 0.01 to 0.5. Each range was discretized into 21 evenly spaced values. These ranges were selected to cover plausible cognitive processes — from weak to strong learning rates, and from highly exploratory to highly deterministic decision-making strategies. To support broader exploration, uniform distributions over the same ranges were used to randomly sample true parameters.

Synthetic behavioral data were generated by initializing agents (RWM or BRM) with randomly sampled parameters. Each agent simulated 100 trials for 11 participants. This was repeated across multiple runs to produce diverse datasets.

Each simulated dataset was then fit using a grid search over the same parameter space. Log-likelihoods were computed for every parameter combination, and the best-fitting parameters were recorded. For RWM, this involved α - β pairs; for BRM, β - λ pairs.

Recovery performance was assessed by correlating true and recovered parameter values using Pearson correlation. Scatter plots were also used to visualize recovery quality, and inter-parameter correlations were examined to check for dependencies in estimation.

Model comparison (& recovery)

Describe how you compared the models.

To compare the models, we computed the *Bayesian Information Criterion* (BIC) for the RWM and the BRM based on their respective maximum log-likelihoods from grid search fitting to the human data. BIC was chosen because it balances model fit with model complexity, penalizing models with more free parameters.

The BIC was calculated using the following formula:

$$\text{BIC} = k \cdot \ln(n) - 2 \cdot \text{LL} \quad (7)$$

where k is the number of free parameters, n is the number of data points (trials), and LL is the maximum log-likelihood.

By comparing BIC values, we assessed which model provided a better trade-off between goodness of fit and complexity. The model with the lower BIC was considered the better explanation of the observed behavior.

how did we perform model recovery.

To evaluate model identifiability and determine whether each model could be reliably recovered from the data it generated, a model recovery procedure was implemented. Each model was initialized using the same parameter values as in the model simulation phase (see section *Model Simulation*), allowing for consistency across analyses.

Using these parameters, each model was used to simulate behavioral data over 100 trials for a fixed number of participants (i.e. 11). The resulting datasets were then fit using both the RWM and BRM, and the log-likelihood of each model was computed.

To assess recovery, we compared the models' log-likelihoods and examined which model yielded the higher log-likelihood when fit to each simulated dataset. This allowed us to determine whether each model was better able to explain its own simulated behavior than that of the competing model.

Successful recovery was defined as a model best fitting the data it generated, indicating that the models are

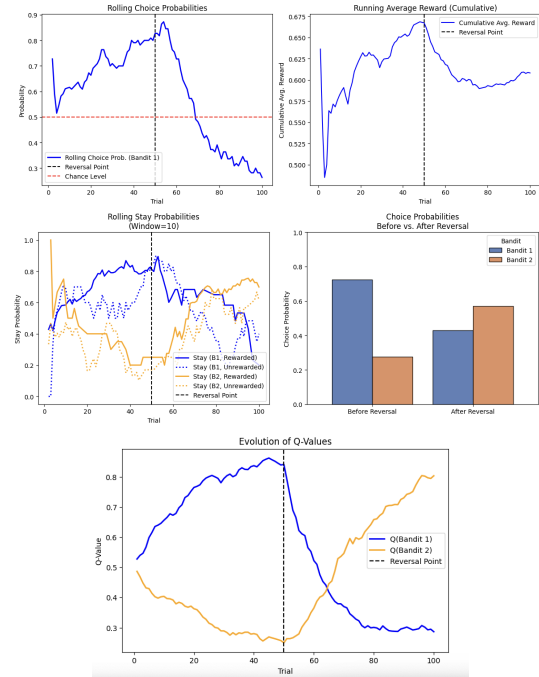
distinguishable given the observed behavior.

Results

Simulation results

Which phenomena do the models capture and why?

Figure 1: Behavioral Plots of the RWM Simulation



The RWM captures gradual, reward-driven adaptation to changes in reward contingencies, as reflected in several behavioral phenomena shown (see Fig. 1).

In the “Rolling Choice Probabilities” plot (top left), the agent shows a clear preference for Bandit 1 before the reversal at trial 50, and then gradually shifts to Bandit 2 — demonstrating learning driven by prediction errors. This slow shift is also reflected in the “Evolution of Q-Values” (bottom), where value estimates for the two bandits cross over as the agent updates its expectations.

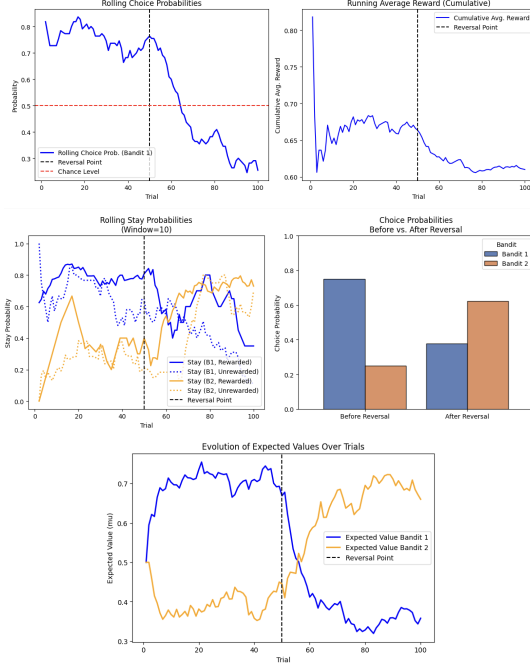
Furthermore, the “Running Average Reward” plot (top right) shows that the model maintains a high reward rate prior to the reversal and slowly recovers performance after the change — consistent with the learning curve of the RWM.

The “Rolling Stay Probabilities” plot (middle left) reveals asymmetric win-stay/lose-shift patterns: the agent tends to repeat choices after reward and switch after non-reward, a hallmark of reward-sensitive learning.

Finally, the “Choice Probabilities Before vs. After Reversal” plot (middle right) confirms the model’s ability to adapt: the preferred bandit changes from Bandit 1 to Bandit 2 after the reversal, reflecting sensitivity to reward structure over time.

Overall, these patterns support the idea that the RWM captures classic reinforcement learning behavior, including reward-driven updating, gradual adaptation, and outcome-sensitive decision-making.

Figure 2: Behavioral Plots of the BRM Simulation



The BRM captures belief-based decision-making and flexible adaptation to environmental change. As shown in Fig. 2, this model reproduces behavior that reflects both reward sensitivity and rapid belief updating.

In the “Rolling Choice Probabilities” plot (top left), agents show a clear preference for Bandit 1 before the reversal and switch quickly to Bandit 2 after trial 50. Unlike the gradual transition in the RWM, this change is more abrupt, indicating that the BRM adapts faster — likely due to the hazard rate promoting belief decay.

This is also seen in the “Evolution of Expected Values” (bottom), where the posterior means (μ) for the bandits sharply invert following the reversal, demonstrating rapid belief updating.

The “Running Average Reward” plot (top right) shows that the Bayesian agent maintains high performance across the task (compared to the RW agent), including a fast recovery after the reversal, indicating efficient adaptation.

The “Rolling Stay Probabilities” (middle left) further highlight reward sensitivity: agents show a strong tendency to stay after reward and switch after non-reward, especially following the reversal.

Finally, the “Choice Probabilities Before vs After Reversal” plot (middle right) shows that the preferred bandit clearly flips from Bandit 1 to Bandit 2 — confirming that the model tracks and responds to structural changes in the

environment.

Together, these results demonstrate that the BRM is well-suited for modeling learning in volatile environments, where agents must adapt quickly based on uncertainty and recent outcomes.

Compared to the RWM, the BRM better captures rapid post-reversal adaptation and belief-based flexibility, whereas RWM captures more gradual, incremental learning.

Which phenomena do the models not capture and why?

The RWM fails to capture rapid shifts in behavior following environmental changes. The transition in choice probabilities and Q-values after the reversal at trial 50 is slow and gradual, rather than immediate (see Fig. 1). This reflects the model’s reliance on incremental error-driven learning without a mechanism for tracking environmental volatility or surprise. As a result, it may underestimate how quickly participants adapt when they detect a change in the reward structure.

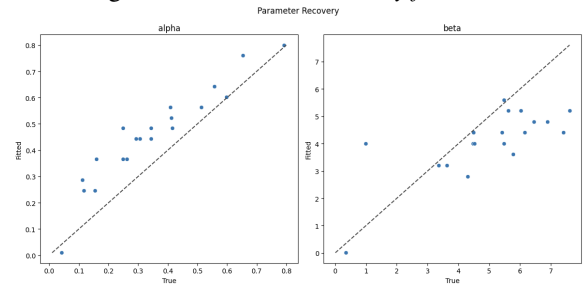
The BRM, while flexible and adaptive, may overestimate how quickly real participants adjust to reversals. The sharp transition in choice probabilities and expected values following the reversal suggests near-instantaneous adaptation (see Fig. 2). This contrasts with human behavior, which often shows some inertia or delay in updating choices. Additionally, the BRM may be too deterministic under strong priors and hazard-rate tuning, potentially failing to capture noise or exploratory deviations often observed in real human data.

While both models capture overall learning dynamics, they do not fully account for occasional lapses or non-systematic choice variability observed in human behavior. Specifically, neither model incorporates mechanisms to address random exploratory choices that are not strictly driven by reward feedback, and they may oversimplify the potential impact of attentional fluctuations or fatigue over 100 trials.

Parameter recovery

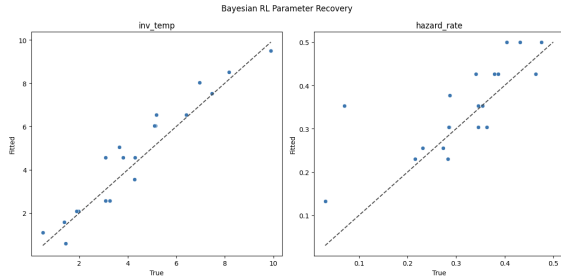
Which parameters can be recovered more reliably, which less reliably?

Figure 3: Parameter Recovery for RWM



As shown in the parameter recovery plot for the RWM

Figure 4: *Parameter Recovery for BRM*



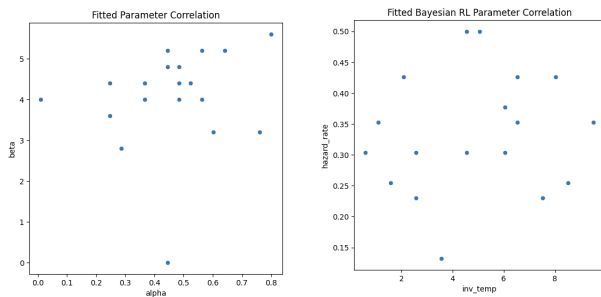
(Fig. 3), both the learning rate (α) and inverse temperature (β) showed significant positive correlations between true and recovered values, indicating that the BRM can recover both parameters reasonably well.

Recovery was stronger for α ($r = 0.83$, $p < .001$), with estimates clustering closely along the identity line, suggesting high precision. In contrast, β recovery was weaker but still significant ($r = 0.61$, $p = 0.004$), showing greater variability—particularly at extreme values.

The correlation between recovered α and β was low and non-significant ($r = 0.19$, $p = 0.41$; Fig. 5, left), suggesting the parameters were not confounded.

In summary, α was recovered more reliably than β , likely due to its direct role in learning, whereas β 's subtler influence on decision noise makes it harder to estimate under noisy conditions.

Figure 5: *Parameter correlations for RWM (left) and BRM (right)*



For the BRM, (Fig. 4), both the inverse temperature (β) and hazard rate (λ) showed strong and statistically significant positive correlations between true and recovered values. The correlation for β was high ($r = 0.90$, $p < 0.001$), with most estimates falling close to the identity line, indicating precise recovery. The hazard rate also showed a strong correlation ($r = 0.76$, $p < 0.001$), although with slightly greater variability, particularly at higher values.

The scatter plot of fitted parameters (Fig. 5, right) revealed no substantial confounding between β and λ , as the correlation between the two recovered parameters was low and non-significant ($r = 0.08$, $p = 0.75$). This suggests that the two parameters were independently recoverable.

Overall, BRM demonstrated strong parameter recoverability, with β being slightly more precise than λ . This supports the interpretability of model fits and confirms that both parameters can be reliably estimated from simulated behavior.

Model recovery

Which model can be recovered more reliably, which less reliably?

Model recovery results showed that the BRM was reliably recoverable when fit to its own simulated data (LL = -662.98 , inverted temperature = 3.07, hazard rate = 0.18), and outperformed the RWM (LL = -667.32 , $\alpha = 0.39$, $\beta = 1.58$) on the same data.

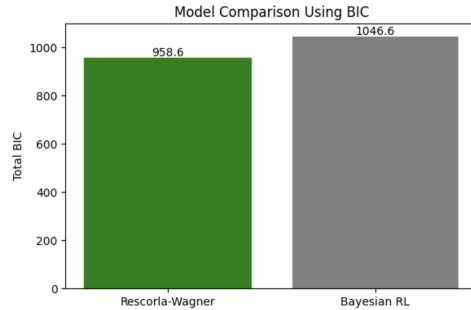
In contrast, when data was generated from RWM, both models fit similarly well (RWM LL = -678.99 , $\alpha = 0.2$, $\beta = 1.58$; BRM LL = -679.69 , inverted temperature = 2.58, hazard rate = 0.08), suggesting that the simpler RWM structure can be mimicked by the BRM.

This indicates that BRM is more distinct and recoverable, while RWM may be harder to uniquely identify.

Model comparison

Which model fits the data better and why?)

Figure 6: *Model comparison using BIC*



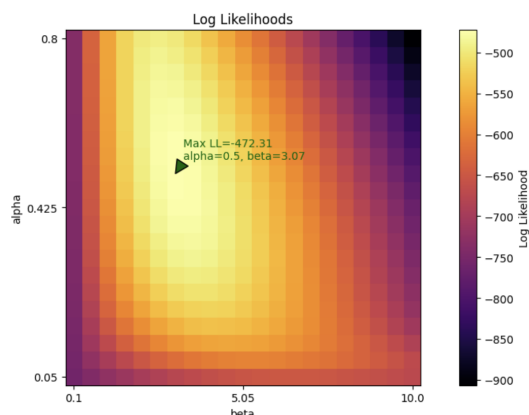
Based on BIC, the RWM provides a better fit to the data (BIC = 958.6) compared to the BRM (BIC = 1046.6). Since lower BIC values indicate better model fit while penalizing for complexity, this large difference ($\Delta BIC \approx 89$) provides strong evidence in favor of the RWM. This suggests that the added flexibility of the BRM does not sufficiently improve fit to justify its complexity in this case.

Parameter fit

Which parameter values fit the data best?

The parameter values of the RWM that best explained the observed behavior were $\alpha = 0.5$ (learning rate) and $\beta = 3.07$ (inverse temperature). These values suggest that participants updated their expectations moderately based on reward outcomes and made choices with a moderate degree of consistency and sensitivity to learned values (see Fig. 5).

Figure 7: *Log-Likelihood Surface for RWM Fit to Human Data*



Discussion

Which hypothesis does our modeling support and why? Based on the model comparison (and model recovery) results.

The modeling supports the hypothesis that human learning in this task can be effectively described by the RW framework, where a fixed learning rate governs prediction error updates. Although the BRM offers adaptive learning through uncertainty tracking, the lower BIC values for the RWM suggest that the simpler error-driven updating mechanism is sufficient to account for the observed behavior. This supports the idea that participants update their beliefs at a consistent rate, even in the face of a reversal, rather than dynamically adjusting based on uncertainty.

Which other insights does our model provide? Based on the parameters fits of the winning model.

The parameter fits indicate that participants exhibit a moderate learning rate, suggesting that they do not completely rely on single trial feedback but rather integrate information over multiple trials. The inverse temperature parameter reflects a balance between randomness and deterministic choice, implying that while participants generally prefer the option with the higher expected value, they also explore alternative options occasionally. This balance may be crucial for adapting to changes in reward contingencies and ensuring flexibility in decision-making.

What are potential weaknesses of your modeling study?

Potential weaknesses include the assumption of homogeneity across participants, as individual differences in attention, risk tolerance, or strategy may not be fully captured by a single fixed parameter set.

The models also do not account for real-world cognitive factors such as lapses in attention or fatigue, which could influence decision-making over 100 trials.

Additionally, the use of grid search for parameter estimation, while systematic, may limit precision compared to more advanced optimization techniques.

Finally, the RWM can be mimicked by the BRM, raising potential identifiability concerns when comparing similar models.

What might be another computational modeling approach for gaining a deeper understanding of the phenomenon?

Another useful modeling approach would be to compare model-free and model-based reinforcement learning strategies. While the RWM and BRM are primarily model-free — relying on learning from past outcomes — model-based approaches incorporate an internal model of the environment, allowing agents to plan ahead or simulate potential outcomes.

Applying a hybrid model that combines both systems could reveal the extent to which participants rely on habitual learning versus goal-directed control, offering a deeper understanding of decision-making processes in dynamic environments (Daw, Gershman, Seymour, Dayan, & Dolan, 2011).

Acknowledgments

This project was completed collaboratively by all group members.

- **Franka Böckmann, Ewa Godlewska and Iheb Marouani:**

Implementation and analysis of the Rescorla-Wagner model, including model simulation & recovery, parameter fitting, and parameter recovery and comparison.

- **David Raul Carranza Navarrete and Julian Calvin Rill:**

Implementation and analysis of the Bayesian reinforcement learning model, including model simulation & recovery, parameter recovery, and comparison.

All authors contributed equally to writing the report, interpreting results, and preparing figures and visualizations.

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